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LLM Assertiveness can be Mechanistically Decomposed into Emotional and Logical Components

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Abstract

Large Language Models (LLMs) often display overconfidence, presenting information with unwarranted certainty in high-stakes contexts. We investigate the internal basis of this behavior via mechanistic interpretability. Using open-sourced Llama 3.2 models fine-tuned on human annotated assertiveness datasets, we extract residual activations across all layers, and compute similarity metrics to localize assertive representations. Our analysis identifies layers most sensitive to assertiveness contrasts and reveals that high-assertive representations decompose into two orthogonal sub-components of emotional and logical clusters-paralleling the dual-route Elaboration Likelihood Model in Psychology. Steering vectors derived from these sub-components show distinct causal effects: emotional vectors broadly influence prediction accuracy, while logical vectors exert more localized effects. These findings provide mechanistic evidence for the multi-component structure of LLM assertiveness and highlight avenues for mitigating overconfident behavior.

1 Introduction

Large Language Models (LLMs) are increasingly deployed in high-stakes domains such as law [10], politics [2], education [16], and public health [15]. A major concern of LLM usage is that they often produce overconfident statements, overstating information regardless of its factual basis [5]. Such behavior can mislead users, amplify bias, and result in poor decisions with serious societal consequences.

Prior work [11, 21, 18] has quantified LLM overconfidence through linguistic assertiveness cues such as "highly certain," or "I am confident that". While prominent in human communication, it remains unclear how these cues are internally represented, and whether LLMs treat assertiveness as a single construct or as multiple, separable components.

Beyond sensitivity to linguistic cues, assertiveness itself plays mixed roles. In human studies, some associate it with certainty and confidence [20, 8], while others link it to persuasion [12]. Distinguishing these roles is critical: if models primarily respond to one, or encode them separately, this implies distinct internal mechanisms.

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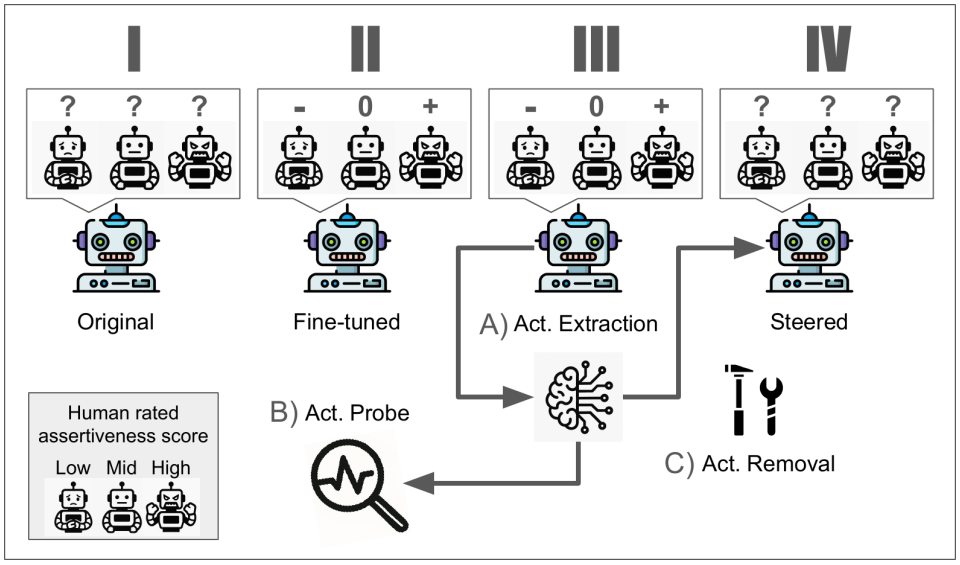

Figure 1: Overview of study design. Phase I: load the base LLM. Phase II: fine-tune on human rated assertiveness scores. Phase III: extract residual activations and cluster samples by similarity to identify assertiveness sub-components (A–B). Phase IV: derive and remove steering vectors for each sub-component and evaluate effects on assertiveness prediction (C). [4]

Mechanistic interpretability provides a way to test this (Figure 1). By analyzing neural activations, we can (A) identify layers and inputs that maximally represent assertiveness, (B) examine how different sub-components interact, and (C) use steering vectors to test their causal influence on model behavior.

Prior theoretical work [4] suggests that related features in neural networks can be encoded as correlated but orthogonal internal representations. For instance, two concepts, such as certainty and persuasion, may often co-occur in language but be stored in distinct, minimally overlapping directions in the model's activation space. This separation allows the network to adjust one without disrupting the other. Yet, this phenomenon has not been systematically tested for socially meaningful traits like assertiveness.

To test these claims, we hypothesize that LLMs represent assertiveness through orthogonal sub-components, notably certainty and persuasion. To evaluate this, we adopt a two-step approach: (1) clustering text samples by activation similarity to uncover latent feature categories (unsupervised), and (2) manipulating their corresponding steering vectors to assess causal effects on model predictions. This approach provides a mechanistic view of how LLMs represent linguistic assertiveness and whether its underlying components can be independently controlled.

2 Methods

2.1 Datasets and model fine-tuning (Phase I-II)

This study used the publicly available assertiveness datasets ($n = 645$) from Ghafouri et al. [5], which were sourced from Anthropic's Persuasion dataset [3], Globe and Mail (GM) Comments [9], Reddit Change My View (CMV) [19], Llama 3–8B-generated arguments on LlAR data [6] and Pei Assertiveness dataset [13]. These datasets contained standardized assertiveness scores rated by expert annotators, with strong inter-annotator agreement ($r > 0.7$) verified by secondary raters.

We fine-tuned a Llama-3.2-1B-Instruct model [6] with Low-Rank Adaptation (LoRA [7]) to predict the assertiveness scores. Datasets were separated into training and validation sets in each fold, one entire dataset (e.g. GM) served as validation (≈ 130 items) and the remaining as training (≈ 515 items) for five folds. Model performance was reported as mean squared error (MSE). After 30 epochs,

[4] Icon credit: Colored bot icon created by Freepik — Flaticon (`https://www.flaticon.com/free-icon/robot_3398567`))

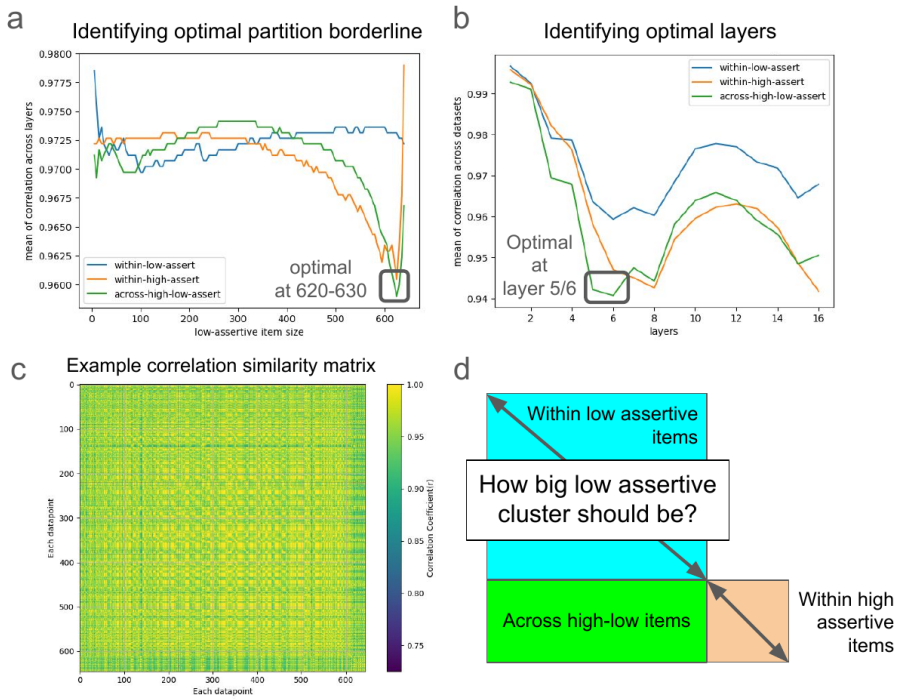

Figure 2: (a) Empirical identification of an optimal partition borderline between low- and high-assertive items; the green curve minimum indicates maximal separation. (b) Layer-wise analysis; the green curve minimum marks the layer with strongest low–high separation. (c) Example correlation similarity matrix across assertive items; brighter values indicate higher correlation. (d) Simplified clusters of (c): within low-assertive items (cyan), across high–low items (green), and within high-assertive items (orange). Each cluster computes the mean correlation of its items.

the mean minimal MSE across the five folds reached 1.94, reducing from 61.11 at the first epoch (See Appendix A.1.1 for details).

2.2 Activation preprocessing and mechanistic interpretability (Phase III-IV)

We fed all samples through the fine-tuned model and extracted per-token activations across all 16 transformer layers, focusing on residual streams at the start of each layer. To reduce dataset-source bias, we removed source-related steering vectors using the Difference-of-Means method [17, 1].

Correlation similarity matrices were then computed across samples, ordered by standardized assertiveness scores (Fig. 2c) and highlighting three key regions: (i) within-low and (ii) within-high assertive clusters and (iii) across high–low pairs (Fig. 2d). By varying the partition borderline between (i) and (ii), we identified the point that minimized cross-cluster similarity, indicating maximal representational separation between low and high assertiveness (Fig. 2a). The layer where this separation peaked was taken as the most sensitive to assertiveness (Fig. 2b).

To probe internal structure, we projected activations of the high-assertive items into a t-SNE map. Spatial groupings were examined to reveal data-driven sub-components (e.g., certainty vs persuasion) for later conceptual interpretation. Steering vectors from each sub-component were selectively removed, and Root Mean Squared Error (RMSE) changes in assertive prediction were measured to test whether these representations function independently or interact (see Appendix A.1.2 for details).

3 Results

Our method identified an optimal borderline separating low- (n = 620) and high-assertive items (n = 25) (Fig. 2a), with middle layers (5–6) exhibiting maximal representational separation (Fig. 2b).

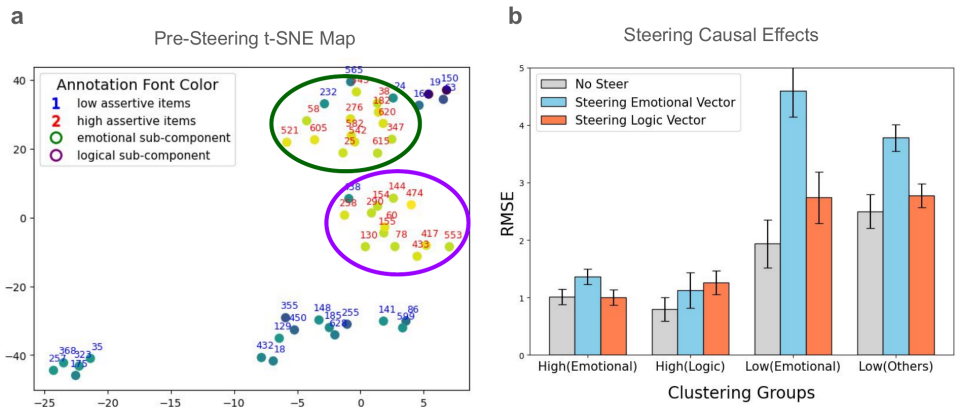

Figure 3: (a) t-SNE visualization of assertiveness clusters, showing low-assertive items (blue indices) and high-assertive items (red indices). Green and purple circles highlight emotional and logical sub-components, respectively (manual labels based on local grouping). Circle fill color indicates standardized assertive score, with lighter shades for higher scores. (b) RMSE changes after removing emotional or logical sub-component vectors, shown separately for high-assertive-logical, high-assertive-emotional, low-assertive-emotional (e.g. index 232, 16, 19), and other low-assertive items (e.g. index 35, 432, 141). Error bars are standard error of the mean (SEM).

With this borderline and layer setting, t-SNE visualizations revealed two distinct clusters within the high-assertive items (n = 25) (Figure 3a). Based on ChatGPT summaries, one cluster was labeled "emotional (big green circle)" and the other "logical (big purple circle)," aligning with the Elaboration Likelihood Model, a long-standing dual persuasion model in Psychology [14]. The theory posits that the logical cluster corresponds to central-route persuasion (evidence, statistics, facts), while the emotional cluster corresponds to peripheral-route persuasion (affective or superficial cues). The logical sub-component partially overlapped with certainty expressions, but which had minor influence on model predictions in these datasets.

Adding extra 25 low-assertive items (blue indices) in the t-SNE map showed partial overlap with the emotional cluster but not the logical one (Figure 3a), suggesting partial independence of the logical sub-component. High-assertive sub-components were spatially cohesive and clearly separating from low-assertive items, reflecting a decision boundary in assertiveness representation space.

Removing the emotional vector broadly degraded performance statistically (i.e. changed beyond SEM error bars), particularly on all low-assertive items and emotionally-relevant high-assertive cluster. In contrast, removing the logical vector primarily affected predictions for logical high-assertive items, indicating a more localized effect (Figure 3b).

4 Conclusion and limitations

We present an empirical method for localizing and interpreting LLM activations linked to assertiveness along a continuous scale. Results show that high-assertive representations split into two orthogonal sub-components—emotional and logical persuasion—each exerting distinct effects when used as steering vectors. Yet, this study has several limitations. The model analyzed is relatively small (1B parameters), leaving scaling effects uncertain, and only one architecture is examined, so cross-model generality remains untested. The dataset size is modest (n = 645), which may obscure finer-grained cues. Moreover, cluster interpretation relies on manual inspection, suggesting that future work should develop more automated approaches. Despite these limitations, this study provides the first visualization of correlated feature decomposition along a continuous assertiveness scale. It offers new methodological insights for analyzing LLM internal representations and contributes to understanding the roots of overconfident behaviors in LLMs.

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